Sublight ion engine Specification

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Revision	B
Status	APPROVED
Classification	CONFIDENTIAL
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Approved by	Cael Merrow
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1. Purpose

This specification defines the performance, interface and verification requirements for the sublight ion engine (PR-SIE) of the Propulsion & Station-Keeping on Starhaven Orbital Habitat & Shipyard.

2. Scope

Fabrication and installation are performed under work package WP-SHV-PR-03 (WBS 1.5.3), owned by Pela Illing.

The prime supplier for flight hardware is Scarif Data Systems. Responsible engineer: Tarn Falkner.

3. Applicable documents

Document	Title
SHV-SYS-SPEC-0001	Starhaven Master System Specification
SHV-PR-DWG-0202	Sublight ion engine general arrangement drawing
SHV-PR-BOM-0302	Sublight ion engine bill of materials
SHV-QA-PLN-0004	Starhaven Verification and Test Plan
SHV-QA-STD-0002	Station Workmanship Standard
SHV-ICD-PD-PR	Interface Control Document: Power Distribution / Propulsion & Station-Keeping

4. Description

Propulsion keeps the station in its assigned orbit and attitude. Station-keeping thrusters and sublight ion engines are commanded by the navigation computer, with attitude control gyroscopes for fine pointing.

The sublight ion engine is delivered as a set of line-replaceable assemblies, integrated on station by Construction Operations and verified by Quality Assurance & Test.

5. Performance requirements

ID	Requirement	Value	Tolerance	Verification
PR-SIE-R01	The sublight ion engine shall provide a engine count of 33 ea.	33 ea	±2%	Analysis
PR-SIE-R02	The sublight ion engine shall provide a thrust per engine of 1,065 MN.	1,065 MN	±2%	Test

5.1 General requirements

ID	Requirement	Verification
PR-SIE-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.	Inspection
PR-SIE-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.	Analysis
PR-SIE-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.	Inspection
PR-SIE-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.	Analysis
PR-SIE-G05	Documentation: Acceptance data packages shall be delivered with every unit.	Inspection

6. Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

7. Verification

Each requirement is verified by the method stated in section 5, following SHV-QA-PLN-0004. Acceptance test reports are filed under the PR test-report series.

8. Revision history

Rev	Date	Description	Author
A	2024-01-04	Initial release.	Tarn Falkner
B	2024-08-31	Editorial corrections.	Tarn Falkner